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MORPHOLOGICAL INTEGRATION IN THE HOMININ DENTITION: EVOLUTIONARY, DEVELOPMENTAL, AND FUNCTIONAL FACTORS

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As the most common and best preserved remains in the fossil record, teeth are central to our understanding of evolution. However, many evolutionary analyses based on dental traits overlook the constraints that limit dental evolution. These constraints are diverse, ranging from developmental interactions between the individual elements of a homologous series (the whole dentition) to functional constraints related to occlusion. This study evaluates morphological integration in the hominin dentition and its effect on dental evolution in an extensive sample of Plio- and Pleistocene hominin teeth using geometric morphometrics and phylogenetic comparative methods. Results reveal that premolars and molars display significant levels of covariation; that integration is stronger in the mandibular dentition than in the maxillary dentition; and that antagonist teeth, especially first molars, are strongly integrated. Results also show an association of morphological integration and evolution. Stasis is observed in elements with strong functional and/or developmental interactions, namely in first molars. Alternatively, directional evolution (and weaker integration) occurs in the elements with marginal roles in occlusion and mastication, probably in response to other direct or indirect selective pressures. This study points to the need to reevaluate hypotheses about hominin evolution based on dental characters, given the complex scenario in which teeth evolve.

KEY WORDS: **Developmental constraints, geometric morphometrics, modularity, merism, serial homology, stabilizing selection.**

Morphological integration is the cohesion among sets of traits that reflects a common influence from functional and/or developmental factors (Klingenberg 2008; Rolian and Willmore 2009). Modularity refers to the relative degrees of connectivity in systems, such that a module is an internally tightly integrated unit relatively independent from other modules (Klingenberg 2008). In practical terms, modules are identified in evolutionary contexts as sets of traits that covary together over evolution, either because they are jointly inherited or selected (Cheverud 1996). These definitions demonstrate that morphological integration and modularity are closely related processes such that modularity can be defined simply as "nested integration" (Willmore et al. 2007).

The study of morphological integration and modularity has a history that can be traced to the middle of the 20th century when the first studies were carried out by Olson and Miller (1958). Since then, researchers have significantly extended the theory and methodology of study (e.g., Cheverud 1996; Raff 1996; Wagner 1996; Rasskin-Gutman 2005; Wagner et al. 2005). In recent years, the number of articles on morphological integration has increased greatly and this topic has been incorporated into the framework of geometric morphometrics with a profusion of publications of both theoretical and empirical studies (e.g., Rohlf and Corti 2000; Hallgrímsson et al. 2004; Klingenberg et al. 2004; Goswami 2006).

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Considerable work has been carried out on the effects of modularity on human evolution, with special emphasis on craniofacial and mandibular morphology (e.g., Bookstein et al. 2003; Lieberman et al. 2004; Bastir et al. 2005; Polanski and Franciscus 2006). As far as dental morphology is concerned, Hlusko and colleagues have used both geometric morphometric and classic morphometric methods to evaluate patterns of integration in hominoid and hominin dentition (Hlusko 2002; Hlusko et al. 2004; Hlusko and Mahaney 2009). These studies have combined morphometric and quantitative genetic data to determine how much of phenotypic correlations between phenotypes result from the genetic correlation between them, in a notable attempt to understand the genetic basis of phenotypic variation (Hlusko 2004). These recent articles have provided a new methodological perspective on classic works, some of which attempted to correlate patterns of variation of different molars in terms of reduction of cusps (e.g., Keene 1965) and size (e.g., Garn et al. 1963).

The dentition as a whole constitutes a developmental module partially independent from its surrounding skeletal parts (see Stock 2001), although the exact degree of genetic and phenotypic independence is still to be clarified (see Butler 1995; Stock 2001; Dayan et al. 2002; Meiri et al. 2005; Miller et al. 2007). At a smaller scale, each individual tooth is a different developmental module because tooth germs can "develop largely independent of the context on which they occur" (Wagner et al. 2005). Developmental and evolutionary observations provide evidence for modularity at both anatomical scales: tooth germs are capable of developing independently of other teeth, even ectopically (Song et al. 2008 and references therein). At the same time, the whole dentition constitutes a relatively independent module, as demonstrated by the loss of the complete dentition in several groups of edentate tetrapods (reviewed in Davit-Béal et al. 2009). This paradox demonstrates that modules are hierarchically structured such that lower level modules are integrated into complexes at a higher level (Wagner 1996). However, these two endpoints offer only an incomplete understanding of the specific factors constraining dental evolution within clades.

Teeth are ideal structures to study modularity due to their serially homologous nature. According to classic work by Bateson (1894) and Butler (1967), teeth are paradigmatic examples of merism, or repeated anatomically homologous structures (see also Kurtén 1953). One of the most important properties of a meristic series is that all of its members are constructed from a common plan, with quantitative differences between elements more than qualitative ones. This meristic array would cause teeth to evolve as part of a system, rather than as individual organs (Townsend et al. 2009). Within this high-level system, a "field" effect can control the differentiation and final shape of teeth into three different dental types: incisors, canines, and molars (Butler 1939), although the "clone model" proposed that each tooth type is intrinsically determined (Osborn 1978). Dahlberg's (1945) adaptation of Butler's concept to the human dentition added a premolar field to the initial three-field paradigm (Townsend et al. 2009), although it has been argued (Butler 1995) that premolars are modified anterior members of the molar field. The lack of canines and premolars in mice—the most commonly used experimental model—has focused studies of dental development on the differentiation between incisors and molars; nonetheless, mouse-based models have been extended to explain how canines and premolars could be produced by overlapping domains of gene expression giving rise to incisors and molars (McCollum and Sharpe 2001). Studies of metameric variation in shape in hominins and hominoids are scarce and based mainly on mandibular molar morphology (Hlusko 2002; Singleton et al. 2011). Nevertheless, Braga and et al. (2010) have evaluated metameric variation at the enamel–dentine junction in the whole postcanine dentition of a small sample that includes the *Australopithecus africanus* fossil Sts52. The extremely reduced sample size used in this study, however, made these authors focus on intraindividual variation (Braga et al. 2010).

The first aim of the present work, then, is to test for the existence of phenotypic modules in the hominin dentition by identifying groups of teeth in which covariation is stronger than between teeth corresponding to different modules. The second aim of this work is to evaluate the effect of morphological integration in the evolution of hominin teeth to ascertain whether evolutionary inferences based on the expectation of neutral evolution are reasonable considering the constraints involved in dental evolution. The evaluation of these aims has crucial importance from a theoretical and empirical point of view. First, the results of this study can help to better understand the effect that morphological integration has on character evolution. Specifically, we aim to evaluate whether morphological integration constrains or facilitates evolution by means of the comparison of the patterns of morphological integration and evolutionary dynamics in the different tooth positions. From an empirical point of view, our results can help to critically reevaluate previously proposed scenarios of hominin evolution if the assumptions of independence and neutral evolution of dental traits are not met, as most phylogenetic reconstructions of the hominin clade rely mainly on craniodental characters. Finally, these results have some implications to understand the evolution of other serially homologous structures, such as vertebrae, ribs, limbs, and digits. These systems evolve under specific conditions related to the common developmental origin of the different elements of a homologous series, so the analysis of dental evolution can provide information on patterns that can be extrapolated to other similar systems.

Table 1. Number of specimens per species and dental class included in the analysis.

	P^3	P^4	M^1	M^2	M^3	P_3	P_4	M_1	M_2	M_3
A. anamensis[1]			2			1				1
A. afarensis	4	3	6	2	2	7	8	6	9	7
A. africanus	7	7	10	9	8	5	2	4	5	8
Paranthropus sp.[2]	7	9	25	7	7	3	6	7	8	7
Undetermined[1,3]	1	4		1	1		3	5	1	4
H. habilis	4	4	10	7	6	5	3	5	4	3
H. ergaster	2	2	3	2		3	3	3	3	2
H. mauritanicus[1,4]			1			1	3		1	2
H. georgicus[5]	1	2	2	2	1	2	2	2	2	1
H. erectus	8	6	5	7	8	10	5	12	11	5
H. antecessor[6]	2	2	3	1		2	4	3	3	2
H. heidelbergensis	17	17	17	19	24	19	22	22	24	23
H. neanderthalensis	16	20	18	18	11	20	18	22	18	14
H. neanderthalensis-H. sapiens[1,7]	1	1	1			1	1			
H. rhodesiensis[1]								1	1	
H. sapiens	46	43	54	45	37	53	44	47	41	36
Total[8,9]	116	120	157	120	105	132	124	139	131	115

[1]Groups not included in the phylogenetically independent analysis, neither in the study of evolutionary modes.

[2]*P. robustus* and *P. boisei* have been pooled together under the term *Paranthropus* sp. due to the less accurate representation of these groups in the sample.

[3]African Plio- and Pleistocene specimens assigned to different species (*Paranthropus* sp., *H. habilis* or *Homo* sp.) by different authors

[4]Lower and Middle Pleistocene specimens from North African sites.

[5]Plio-Pleistocene fossils from Dmanisi site (Georgia).

[6]Lower Pleistocene fossils from Atapuerca-TD6 site (Spain).

[7]Neanderthal or modern human remains without clearly diagnostic features or without published specific assignment.

[8]Only teeth belonging to the same individuals have been included in pairwise comparisons.

[9]Mean shapes of each species used for independent contrast analysis have been calculated using all the specimens for each dental class and species.

Material and Methods

MATERIAL

The study sample (Table 1) includes teeth belonging to all postcanine dental classes and to the majority of species of the genera *Australopithecus, Paranthropus* and *Homo* (detailed descriptions of the composition of these samples can be found in Gómez-Robles 2010). Anterior teeth have not been included in these comparisons due to methodological problems to accurately describe their morphology with two-dimensional geometric morphometric coordinates (Gómez-Robles 2010).

All analyses were carried out using photographs of the occlusal surface of teeth. A Nikon D1H digital camera (Tokyo, Japan) fitted with an A/F Micro-Nikkor 105 mm, f/2.8D, was used to collect the images and the depth of field was maximized by adjusting it to f/32. Attachment to a Kaiser Copy Stand Kit RS-1 (Buchen, Germany) ensured uniform positioning of the camera. Each tooth was positioned so that the plane corresponding to the cemento–enamel junction and the occlusal surface was parallel to the lens of the camera. Further details regarding the positioning of teeth and the error introduced during the photography process are described by Gómez-Robles et al. (2008). Right lower teeth and left upper teeth were selected for study when both antimeres were present, and opposite antimeres were mirror-imaged when the first one was absent, broken, or the identification of landmarks was unclear. This strategy to increase sample size also implies that in some comparisons, teeth belonging to different sides of the same individual were used without controlling for developmental noise causing fluctuating asymmetry (see Laffont et al. 2009).

GEOMETRIC MORPHOMETRICS

Different conformations of landmarks and semilandmarks were used to describe dental morphology. These conformations included from four up to eight landmarks (depending on the studied tooth) located at cusp tips and groove intersections (Fig. S1–S10). They also included from 30 up to 40 sliding semilandmarks (Bookstein 1996, 1997) to describe the dental periphery. The detailed conformations of landmarks and semilandmarks, as well as their definitions, are provided in the supplementary on-line information. After semilandmarks were slid such that the Procrustes distance between conformations was minimized (see Pérez et al. 2006), they were treated as geometrically homologous landmarks. Generalized Procrustes analysis (Rohlf and Slice 1990) was used

to extract all of the geometric information remaining in the sample after translating, scaling, and rotating the landmark configurations so that the distances between corresponding landmarks are minimized following a least-squares criterion. Relative warp analysis (Bookstein 1991), which captures the main patterns of morphological variation, was used as a previous step for some subsequent analyses.

EIGENVALUE DISTRIBUTION

Differences in the distribution of variance have been proposed to be a proxy to the degree of morphological integration of biological structures (Wagner 1984). Integration within the different teeth was evaluated accordingly by evaluating the dispersion of the eigenvalues in the different tooth positions. Significant differences in eigenvalue distributions were tested by comparing the variance of the eigenvalues (EV) of 1000 bootstrapped samples for each dental class (e.g., Wagner 1984; Young 2006; Pavlicev et al. 2009). Since it is calculated from covariance matrices, EV is scale-dependent, so eigenvalues were standardized by the total shape variance (Young 2006). EV also depends upon the number of traits (Pavlicev et al. 2009), so it was calculated by using only those relative warps that account for 99% of total Procrustes variance of each sample (Gómez-Robles et al. 2011) to circumvent the existence of relative warps accounting for almost no variance (due to the large number of semilandmarks). Calculations were carried out with *Mathematica 8.0* (Wolfram Research Inc., Champaign, IL, USA) using routines written by Polly and Goswami (see Goswami and Polly 2010a) and available online (http://mypage.iu.edu/~pdpolly/Software.html).

INTERTOOTH COMPARISONS

From among the different methods available for studying morphological integration in the complete dentition (Goswami and Polly 2010a), the two-block partial least-squares analysis (2B-PLS, Rohlf and Corti 2000; Bookstein et al. 2003) and the R_V coefficient analysis (Klingenberg 2009) were used in this study. 2B-PLS method compares two morphological sets by using a singular value decomposition of the cross-covariance matrix, finding new pairs of axes (called singular axes or singular warps) that account for the maximum amount of covariance between both sets. This analysis has been used to extract the main trajectories of covariation in each pairwise comparison. The vector correlation coefficient (R_v, Escoufier 1973) has been used to calculate the multivariate correlation between the two blocks (Klingenberg 2009). This coefficient is a multivariate analogue to a squared correlation coefficient R^2. It has a maximum value of 1 when all the variation can be predicted across blocks and a minimum value of 0 when complete modularity exists. The present study has evaluated pairwise correlations between all tooth positions (see Laffont et al. 2009; Renaud et al. 2009) using MorphoJ software (Klingenberg 2011). For those teeth that are not located in situ in their corresponding maxilla or mandible, published individual associations (Lumley et al. 1972; Wolpoff 1979; Bermúdez de Castro et al. 1999, 2004, 2006; M.A. Lumley and A. Vialet, pers. comm.) were used to include teeth belonging to the same individuals in pairwise comparisons.

The correlated evolution of postcanine teeth was also evaluated after phylogenetic effects were removed (Arnqvist and Rowe 2002). PHYLIP software (Felsenstein 2005) was used to calculate phylogenetically corrected scores (calculated as independent contrasts of relative warp scores obtained in the analyses of species mean shapes). These scores were used to calculate a phylogenetically corrected R_V coefficient (Drake and Klingeneberg 2010). Independent contrasts (Felsenstein 1985) were calculated using a phylogeny based on some modifications of reviews by Wood and Richmond (2000) and Wood and Lonergan (2008), and branch lengths were included in these calculations (Fig. 1).

The independent contrasts approach, however, has two main shortcomings (apart from the uncertainty about the topology of the human phylogeny). Firstly, it is based on a Brownian motion model of evolution (Felsenstein 1985), which may not be true in the case of human teeth (see below). Secondly, the estimates of the mean or consensus morphologies of the different species have large standard errors when sample sizes are small. Some of the species in this study are known from only one or two specimens, so mean values in these cases are unlikely to be an accurate representation of the species mean shape. It has been also claimed that the results of across-species and phylogenetically corrected analysis should not be reported simultaneously because they rely on very different assumptions (Freckleton 2009). In the present study, across-species comparisons have been carried out to evaluate integration without regard to the uncertain matter of taxonomic assignment and reconstruction of the hominin phylogeny, but special attention has been paid to phylogenetically corrected comparisons.

COMPARISON OF INTER- AND INTRASPECIFIC PATTERNS OF INTEGRATION

The effects of modularity have been tested both in an interspecific and intraspecific context. The intraspecific evaluation has been based on a sample of *H. sapiens* teeth, because this is the best represented species. Comparison of the results obtained in the two different contexts is likely to provide information about the hierarchical nature of morphological integration in terms of similar or different patterns of covariation at the level of species and clades (Fig. 2, Claude 2004). It should be noted that the adjective *hierarchical* is used in this article with two different meanings. First, it is used to make reference to the existence of modules within other modules (as mentioned in the Introduction). Second, the

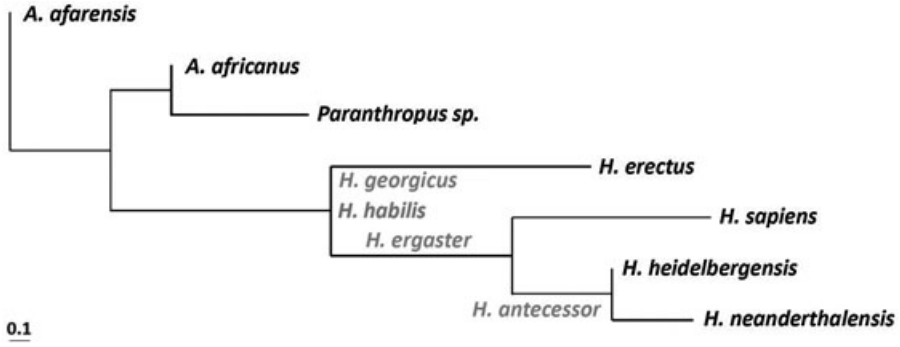


Figure 1. Phylogeny employed in independent contrasts analysis and in the study of the mode of evolution of teeth. Branch lengths represent millions of years. Chronologies of species spanning a long temporal period have been averaged using published datings for these species and specific chronologies of the fossil samples included in this study (but see Hunt 2004). Species represented in gray are those whose phylogenetic position is less clear. These species have been excluded from the study of evolutionary modes.

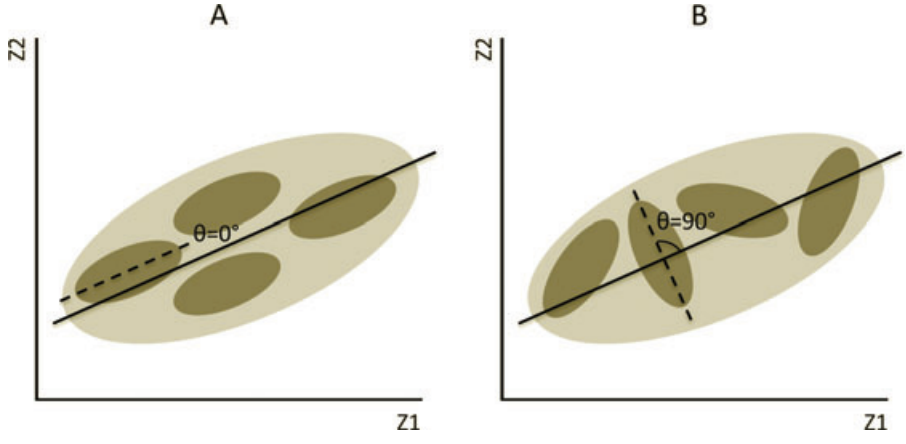


Figure 2. Comparison of morphological integration at different hierarchical levels: within species (small dark ellipses) and among species (large light ellipses). The ellipses illustrate the variance–covariance between two traits (although the same abstraction can be extrapolated to multivariate data, Steppan et al. 2002). (A) Similar patterns of intra- and interspecific morphological integration; singular vectors form an angle θ close to 0°. (B) Intraspecific patterns of morphological integration differ among different species and with the interspecific observation; singular vectors form an angle θ close to 90°. Note that this figure would represent PLS scores for the comparison of two morphological traits, not the singular vectors from which these scores are calculated. Modified after Claude (2004) and Hunt (2007a).

hierarchical nature of morphological integration evaluated here corresponds to the comparison of patterns of integration observed at different taxonomical levels (among species vs. within species).

The similarity of the main covariation trajectories has been measured as the angle between the inter- and intraspecific first singular vectors in each pairwise comparison. These trajectories can be considered significantly correlated when the angle between the two vectors—calculated as the arccosine of the inner product of the two vectors after both are scaled to unit length (Hunt 2007a)—is outside the confidence interval of the angle between two randomly selected vectors (1000 pairs of random vectors) of the same length as the ones being compared (Hunt 2007a; Renaud et al. 2009).

EVOLUTIONARY MODES

Evolutionary modes of the different teeth have been analyzed by studying the scaling relationship of shape divergence and time since common ancestry (Gingerich 1993; Polly 2001, 2004, 2008). Hence, evolutionary modes were estimated by means of the equation $y = x^a$, where y is the shape difference between two taxa (measured as the Procrustes distance between the mean shapes of both taxa), x is the time elapsed since their last common ancestor, and a is the coefficient that determines the type of phenotypic evolution. The exponent a ranges from 0 to 1. An exponent of 0 corresponds to stasis (phenotypic divergence does not increase with time), whereas an exponent of 1 corresponds to directional evolution (morphological divergence increases linearly with time). A value of 0.5 corresponds to neutral evolution following a typical Brownian motion model (divergence increases with the squared root of time), and intermediate values are indicative of random evolution with predominance of stasis ($a < 0.5$) or directional evolution ($a > 0.5$). The value of a was calculated by fitting the equation $y = x^a$ to the data with exponents ranging from 0.1 to 1 at 0.1 intervals. The value that minimized the residual

variance was chosen as the one that best describes the relationship between morphogical and phylogenetic divergence. A randomization procedure with 1000 random permutations has been used to calculate confidence intervals for the value of a using *Mathematica 8.0*.

Divergence time, the sum of the time elapsed between the occurrences of two taxa and their last common ancestor, was estimated using the paleontological and archaeological records. These estimates, however, are unavoidably affected by the uncertainty about the exact chronology and phylogenetic relationships of some of the species included in comparisons. Moreover, these uncertainties affect not only to the species whose phylogenetic position and/or chronology is not clear, but also to all the species connected through an uncertain node (Polly 2008). For this reason, the employed phylogeny has excluded those species whose phylogenetic position is more controversial to minimize the error introduced in the comparisons. The species included in this analysis after these considerations were taken into account are *A. afarensis*, *A. africanus*, *Paranthropus* sp., *H. erectus*, *H. heidelbergensis*, *H. neanderthalensis*, and *H. sapiens* (see Fig. 1).

Results

INTEGRATION WITHIN TEETH

The evaluation of the eigenvalue variance reveals that there are significant differences in integration among the different tooth positions ($F = 16{,}601.9$; $P << 0.001$). Post-hoc tests demonstrate that differences in EV are significant between every pair of tooth classes. The box plot corresponding to resampled EVs for upper and lower postcanine teeth (Fig. 3, inset) shows a clear pattern in the distribution of this variance. In both the upper and lower dentitions, first molars have the lowest EV ($EV_{UM1} = 0.0039$ and $EV_{LM1} = 0.0021$), and this variance increases gradually toward premolars and toward distal molars. Lower premolars have higher EVs than upper premolars ($EV_{LP3} = 0.0076$ and $EV_{LP4} = 0.0079$ vs. $EV_{UP3} = 0.0063$ and $EV_{UP4} = 0.0055$), whereas upper molars have higher EVs than lower molars ($EV_{UM1} = 0.0039$; $EV_{UM2} = 0.0068$; and $EV_{UM3} = 0.0112$ vs. $EV_{LM1} = 0.0021$; $EV_{LM2} = 0.0030$; and $EV_{LM3} = 0.0040$). These values indicate strong integration in upper third molars and lower first premolars and weak integration in first molars. These results also indicate stronger integration in lower than upper premolars and in upper molars than lower molars.

INTERSPECIFIC SHAPE COVARIATION

If covariation between postcanine teeth is evaluated without regard to the specific assignment of individuals, almost all the results show significant covariation, with R_V coefficients ranging from 0.20 to 0.53 (Table 2). These values are substantially higher than the ones observed in other mammalian groups, such as voles ($R_V = 0.17$–0.23, Laffont et al. 2009). However, numerical values are only of limited biological relevance, mainly because of differences in sample sizes and conformations of landmarks.

Phylogenetically corrected R_V coefficients (Table 3) are high, sometimes very high, with values ranging from 0.54 to 0.89. As these comparisons are based on species mean shapes, sample sizes are generally the same (with only a few exceptions), so numerical values in this case are biologically more comparable. These values are considerably higher than those corresponding to among individual comparisons, both in this study and in other published studies, and they reveal a strong evolutionary integration in the complete postcanine dentition in hominins. Within this pattern of general integration, some combinations of teeth have higher R_V coefficients indicating stronger integration. Integration is generally stronger in the lower dentition (mean $R_V = 0.690$) than in the upper dentition (mean $R_V = 0.669$), and also between molars (mean $R_V = 0.662$ in upper molars and mean $R_V = 0.729$ in lower molars) than between premolars ($R_V = 0.574$ in upper premolars and $R_V = 0.655$ in lower premolars), in both the upper and the lower rows. Cross-comparisons between premolars and molars reveal similar degrees of covariation to those observed between premolars and between molars in both the maxillary and the mandibular dentitions (mean $R_V = 0.688$ in the upper dentition and mean $R_V = 0.676$ in the lower dentition). As for antagonist teeth, the highest covariation is observed between first molars. R_V coefficients decrease gradually toward the most mesial and most distal elements of both arcades (Table 3 and Fig. 4). This pattern of decreasing covariation gives rise to lower R_V coefficients between first premolars and third molars, which are not significant at the significance threshold of 0.01.

INTRASPECIFIC SHAPE COVARIATION

Intraspecific comparisons reveal a similar (although non identical) pattern than that observed in interspecific comparisons (Fig. 5). In the upper dentition, highly significant, significant, or almost significant covariation (considered at the 0.01, 0.05, or 0.1 levels, respectively) is observed between the first premolar and the second premolar, first and second molars (Table 4). Significant covariation is also observed between the first upper molar and the second and third upper molars. A highly significant R_V coefficient is observed between both lower premolars ($R_V = 0.284$; $P < 0.001$), and significant or almost significant coefficients correspond to the comparisons between the lower second premolar and lower first and second molars. As for antagonist teeth, only first molars show highly significant covariation within *H. sapiens* ($R_V = 0.350$; $P = 0.003$), although first premolars show almost significant covariation ($R_V = 0.205$; $P = 0.096$).

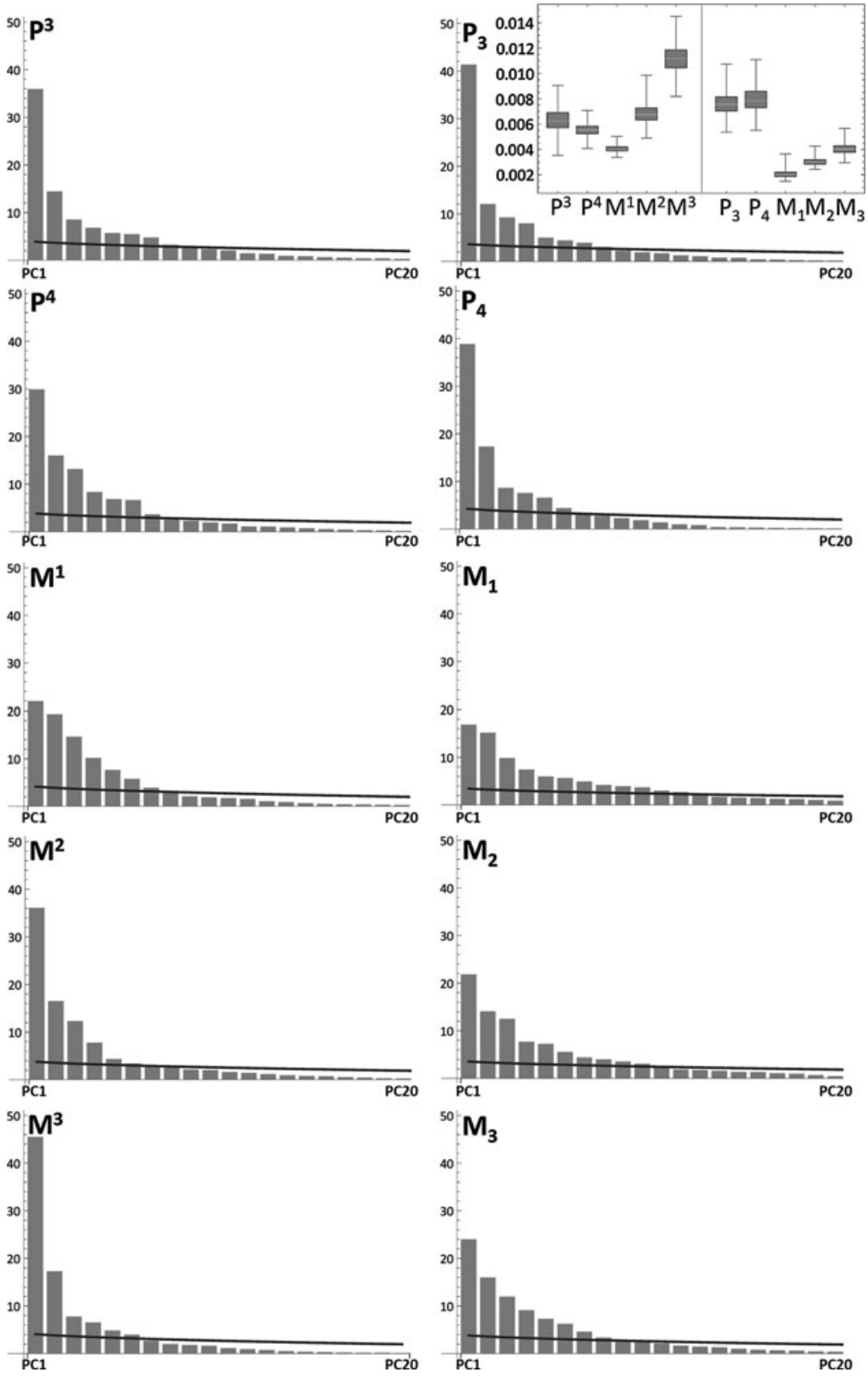


Figure 3. Scree plots representing the distribution of variance in the different teeth against a random model of no integration (solid lines). Random models correspond to broken stick distributions with eigenvalues obtained from random matrices of the same dimensions than the studied covariance matrices. Only the first 20 principal components are represented to facilitate visualization. Inset: Box plot corresponding to the comparison of eigenvalue variance in hominin postcanine teeth. Left half: upper dentition. Right half: lower dentition. Comparisons based on 1000 bootstrapped samples.

Table 2. R_V coefficients obtained through across-species comparisons.

	P^3	P^4	M^1	M^2	M^3	
P_3	0.226*** $P < 0.001$ $N = 53$	0.288*** $P < 0.001$ $N = 77$	0.303*** $P < 0.001$ $N = 68$	0.218*** $P < 0.001$ $N = 72$	0.271*** $P = 0.001$ $N = 43$	P^3
P_4	0.346*** $P < 0.001$ $N = 82$	0.288*** $P < 0.001$ $N = 47$	0.306*** $P < 0.001$ $N = 72$	0.330*** $P < 0.001$ $N = 80$	0.396*** $P < 0.001$ $N = 46$	P^4
M_1	0.215*** $P < 0.001$ $N = 75$	0.268*** $P < 0.001$ $N = 71$	0.290*** $P < 0.001$ $N = 56$	0.254*** $P < 0.001$ $N = 71$	0.453*** $P < 0.001$ $N = 41$	M^1
M_2	0.328*** $P < 0.001$ $N = 81$	0.263*** $P <0.001$ $N = 80$	0.346*** $P < 0.001$ $N = 85$	0.283*** $P < 0.001$ $N = 51$	0.533*** $P < 0.001$ $N = 51$	M^2
M_3	0.307*** $P < 0.001$ $N = 51$	0.202*** $P = 0.008$ $N = 49$	0.240** $P = 0.010$ $N = 48$	0.385*** $P < 0.001$ $N = 54$	0.312* $P = 0.080$ $N = 22$	M^3
	P_3	P_4	M_1	M_2	M_3	

*$P < 0.1$; ** $P < 0.05$; ***$P < 0.01$.

N: number of individuals included in each pairwise comparison.

The gray-shaded diagonal represents comparisons between antagonist teeth. Upper dentition above the diagonal and lower dentition below the diagonal.

Table 3. Phylogenetically corrected R_V coefficients.

	P^3	P^4	M^1	M^2	M^3	
P_3	0.571* $P = 0.015$	0.574* $P = 0.026$	0.712** $P = 0.007$	0.543* $P = 0.028$	0.826** $P = 0.001$	P^3
P_4	0.655** $P = 0.002$	0.813** $P = 0.001$	0.709** $P = 0.006$	0.751** $P = 0.001$	0.589* $P = 0.041$	P^4
M_1	0.798** $P < 0.001$	0.686** $P = 0.004$	0.895** $P = 0.001$	0.739** $P = 0.002$	0.641* $P = 0.023$	M^1
M_2	0.827** $P < 0.001$	0.619** $P = 0.006$	0.757** $P = 0.001$	0.754** $P < 0.001$	0.607* $P = 0.041$	M^2
M_3	0.585* $P = 0.010$	0.539* $P = 0.041$	0.797** $P = 0.001$	0.634** $P = 0.008$	0.611* $P = 0.034$	M^3
	P_3	P_4	M_1	M_2	M_3	

*$P < 0.05$; **$P < 0.01$.

Number of individuals used to estimate the mean shapes from which contrasts have been calculated is provided in Table 1.

The gray-shaded diagonal represents comparisons between antagonist teeth. Upper dentition above the diagonal and lower dentition below the diagonal.

COMPARISON OF INTRA- AND INTERSPECIFIC TRAJECTORIES OF COVARIATION

The first axes of covariation between different postcanine teeth tend to show correlated distal reductions between teeth located in the upper row, in the lower row, and between antagonist teeth (Figs. S11–S16). These correlated reductions of the distal areas of postcanine teeth are observed not only between premolars (Fig. S13) or between molars (Figs. S12 and S15), but also in premolar–molar cross-comparisons (Figs. S11 and S14). Intraspecific main axes of covariation tend to involve also a reduction of the distal areas. However, intraspecific ranges of variation are much more reduced than interspecific ranges, so some of these distal reductions are more difficult to visualize in the grids corresponding to *H. sapiens* comparisons (Figs. S17–S21). Nonetheless, the quantitative comparison of the main axes of covariation demonstrates that these axes are overall conserved both within and among species in those cases where significant correlations are observed within *H. sapiens*. In the majority of the comparisons, the intra- and interspecific first singular vectors form angles ranging from 30° to 60° (Table 5). As the significance threshold

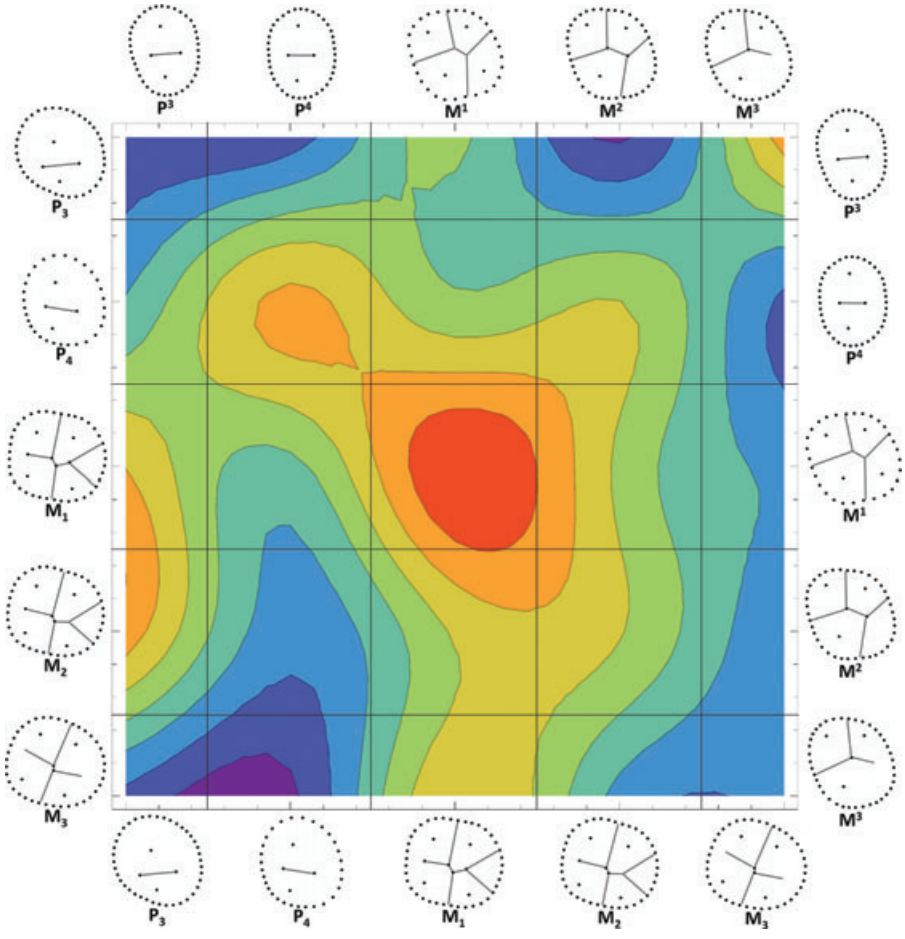


Figure 4. Contour line diagrams showing correlation fields (Kurtén 1953) in the hominin postcanine dentition. Red represents strong integration and blue represents weak integration. The upper right half of the graph corresponds to the upper dentition, the lower left half, to the lower dentition, and the diagonal, to the correlations between antagonist teeth (as in Tables 2–4). TPS grids correspond to the mean shape of each tooth. Mesial margins are represented to the left, distal margins to the right, buccal margins to the top, and lingual margins to the bottom of the figure. Dental grooves have been drawn manually based on the observed morphological patterns in the areas without relevant landmarks to facilitate visualization.

for correlated vectors of this length has been established in approximately 75°, this quantitative comparison demonstrates the general maintenance of these covariation patterns, even if they are not easy to ascertain visually by means of thin plate spline (TPS) grids.

EVOLUTIONARY MODES

The scaling relationship between morphological divergence (Procrustes distance) and phylogenetic distance (the sum of time since common ancestry for pairs of species) reveals differences in the evolutionary mode at the diverse tooth positions (Fig. 6). In both the upper and lower arcades, first molars show the lowest scaling coefficients ($x^{0.2}$ in lower first molars and $x^{0.3}$ in upper first molars), which indicate highly constrained morphological divergence. This coefficient increases toward the most mesial and most distal elements of both arcades, giving rise to best fits at $x^{0.6}$ in lower first premolars and $x^{0.7}$ in upper third molars. These values point to a combination of random evolution with slight directional trends in the morphological evolution of these teeth. The observed values are in general lower in the mandibular dentition than in the maxillary dentition, the scaling values of which point to a more neutral mode of evolution.

Discussion

Much of our recent evolutionary history has been inferred using teeth because they are the most common and best-preserved organs in fossils (Tucker and Sharpe 2004). A clear understanding of the way teeth evolve and of the factors impacting their morphological change is then crucial to make correct inferences. Several articles have demonstrated that cranial evolution in hominins can be best described by a neutral pattern in the evolution of the genus *Homo* (Ackermann and Cheverud 2004), in the

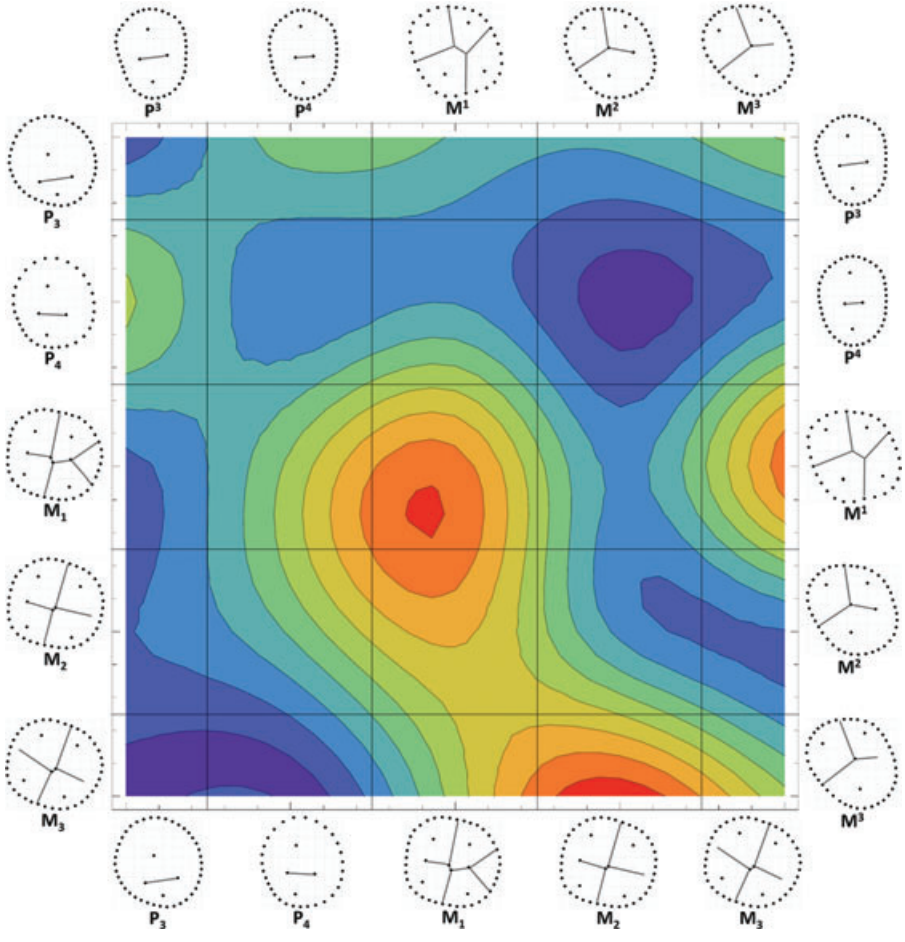


Figure 5. Contour line diagrams showing correlation fields in the *H. sapiens* postcanine dentition. Red represents strong integration and blue represents lack of integration. TPS grids correspond to the *H. sapiens* mean shape for each tooth. Same conventions as in Figure 4.

divergence between Neanderthals and modern humans (Weaver et al. 2007) and in the diversification of *H. sapiens* (e.g., von Cramon-Taubadel 2009; Betti et al. 2010). The same expectation of neutral evolution has been assumed explicitly or implicitly in some recent studies for dental traits (Martinón-Torres et al. 2007; Bailey et al. 2009) on the basis of an expected similarity between cranial and dental characters. Although this is a necessary assumption in some cases, it has been demonstrated that cranial and dental features may be subject to different selective pressures, being effectively located in different evolutionary scenarios (Dayan et al. 2002). It is important to note that strong departures from a random walk mode of evolution may alter the expected patterns of morphological similarities and divergences, thus biasing the inferred evolutionary scenarios and decreasing the ability to correctly classify hominin specimens and species. However, it is also worth noting that, when all the complexity of the evolutionary process is not accurately reflected in a model, Brownian motion models can outperform more complex ones that lack some relevant information or that include wrong information about evolutionary parameters (Butler and King 2004).

INTEGRATION WITHIN TEETH

Eigenvalue variance evaluations reveal the highest integration at the most variable teeth (upper third molars and lower first premolars) and, conversely, the lowest EVs within each arcade correspond to the most stable teeth, namely first molars. Following Wagner's (1984) reasoning, the most variable teeth are also the most integrated, whereas the most stable teeth are the least integrated. A similar result (strong integration linked to high morphological disparity) has been observed in the molar region of Carnivora and has been explained as a result of a strong selective pressure on this region (Goswami and Polly 2010b). The association between high morphological integration and high variability stems from the necessity of highly variable structures to be strongly integrated to keep their structural correlations and, hence, their functionality. An example of this is observed in the different levels of

Table 4. R_V coefficients obtained in the intraspecific analysis of *H. sapiens*.

	P^3	P^4	M^1	M^2	M^3	
P_3	0.205* $P = 0.096$ $N = 37$	0.271* $P = 0.063$ $N = 34$	0.258** $P = 0.027$ $N = 37$	0.246** $P = 0.036$ $N = 34$	0.282 $P = 0.439$ $N = 21$	P^3
P_4	0.284*** $P < 0.001$ $N = 40$	0.224 $P = 0.200$ $N = 29$	0.239 $P = 0.120$ $N = 38$	0.191 $P = 0.216$ $N = 36$	0.229 $P = 0.954$ $N = 19$	P^4
M_1	0.213 $P = 0.170$ $N = 36$	0.285* $P = 0.054$ $N = 29$	0.350*** $P = 0.003$ $N = 34$	0.235** $P = 0.037$ $N = 35$	0.352** $P = 0.035$ $N = 21$	M^1
M_2	0.219 $P = 0.275$ $N = 34$	0.263* $P = 0.081$ $N = 30$	0.325** $P = 0.016$ $N = 30$	0.235 $P = 0.200$ $N = 31$	0.219 $P = 0.533$ $N = 19$	M^2
M_3	0.188 $P = 0.951$ $N = 22$	0.184 $P = 0.765$ $N = 21$	0.289 $P = 0.266$ $N = 22$	0.374 $P = 0.277$ $N = 17$	0.286 $P = 0.547$ $N = 15$	M^3
	P_3	P_4	M_1	M_2	M_3	

*$P < 0.1$; **$P < 0.05$; ***$P < 0.01$.

N: number the individuals included in each pairwise comparison.

The gray-shaded diagonal represents comparisons between antagonist teeth. Upper dentition above the diagonal and lower dentition below the diagonal.

[Correction made to Table 4 after initial online publication February 23, 2012.]

Table 5. Angles between inter- and intraspecific first singular vectors.

Comparison[1]	Angle 1[2]	Significance threshold 1[2,3]	Angle 2[2]	Significance threshold 2[2,3]
P^3–P^4	52.68°	77.37°	52.91°	77.37°
P^3–M^1	33.68°	77.37°	62.69°	76.15°
P^3–M^2	34.47°	77.37°	41.75°	77.66°
M^1–M^2	67.69°	76.15°	40.25°	77.69°
M^1–M^3	39.34°	76.15°	40.64°	77.52°
P_3–P_4	36.28°	77.10°	31.13°	76.15°
P_4–M_1	30.86°	76.15°	43.14°	78.51°
P_4–M_2	64.96°	76.15°	77.89°	78.05°
M_1–M_2	45.34°	78.51°	31.38°	78.05°
P^3–P_3	33.97°	77.37°	22.55°	77.10°
M^1–M_1	60.46°	76.15°	53.88°	78.51°

[1]Angles have been measured only in those pairwise comparisons where significant correlations are found in intraspecific analyses.

[2]Angle 1 and Significance threshold 1 correspond to the first tooth specified in the first column. Angle 2 and Significance threshold 2 correspond to the second tooth.

[3]Angles below the significance threshold correspond to significantly correlated singular vectors (0°: parallel vectors; 90°: orthogonal vectors).

integration corresponding to upper and lower third molars. The lower level of integration of mandibular third molars can be linked to their random pattern of reduction, with frequent appearance of secondary cusps and crenulations concomitant to the loss of main cusps. Alternatively, highly integrated upper third molars evince a consistent pattern of reduction, at least in the studied populations: the hypocone is reduced and lost first and the metacone second, whereas the protocone and paracone keep a more constant size, consistent with the findings that mesial cusps are more conservative (Corruccini 1979; Kondo and Yamada 2003; Takahashi et al. 2007).

Differences in the degree of integration are also related to the mode of evolution, as demonstrated by the matching between EV values and the type of selection (low EV is associated with stasis and high EV with directional evolution). One explanation for this pattern can be found in the idea of evolution following the lines of least resistance (Schluter 1996) and in the concept of adaptive landscape (Wright 1932). The small eigenvalue

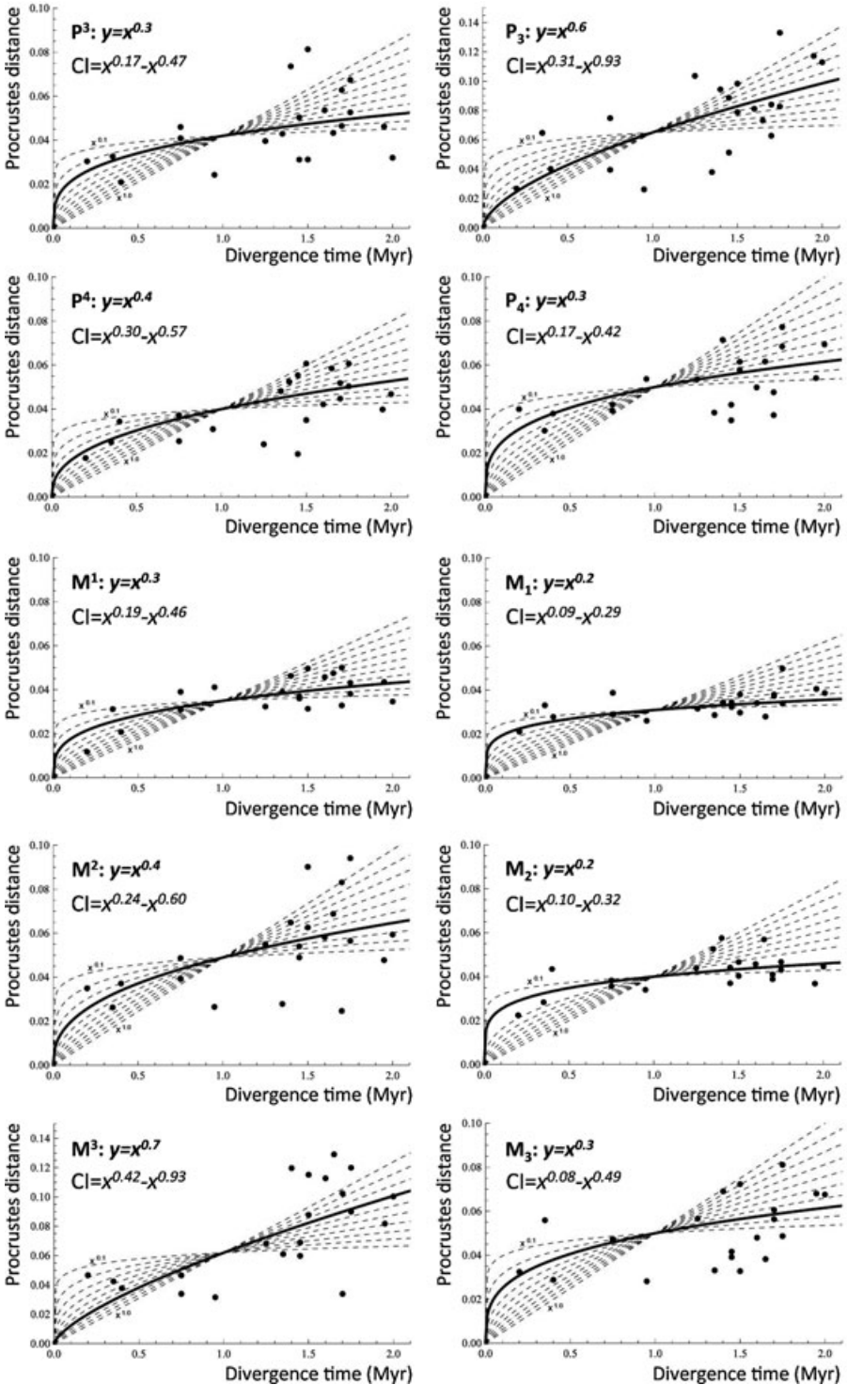


Figure 6. Mode of evolution of postcanine teeth inferred by fitting the equation $y = x^a$ (where x represents time since common ancestry and y represents phenotypic divergence measured as Procrustes distance). Upper dentition represented to the left and lower dentition to the right. $x^{0.1}$ corresponds to stasis; $x^{0.5}$ corresponds to neutral evolution; x^1 corresponds to directional evolution. Intermediate values as the ones observed in this figure correspond to neutral evolution with predominance of stasis ($a < 0.5$) or directional change ($a > 0.5$). Confidence intervals based on 1000 randomizations are provided below the best-fit equations.

variance observed in first molars could be considered to be an adaptive character under this formalism if a correlation between the genetic and phenotypic covariance matrices is assumed (e.g., Oliveira et al. 2009), even though accepting the complexity of the genotype–phenotype mapping (see Pigliucci 2010). Because each eigenvector corresponds to a directional axis in multivariate phenotypic space, a low EV would correspond to a situation where many eigenvectors are associated to similar amounts of variance without a clearly predominant direction. In this case, those populations whose mean value departs from the selective optimum (due to random processes such as genetic drift) would have different directions through which the optimum can be reached again (Steppan et al. 2002). This scenario provides a mechanism through which first molar morphology might remain stable throughout time, because the optimum morphology can be easily recovered in any direction. On the contrary, when the majority of variance is confined to one eigenvector, the remaining ones will be associated with no or little variance, all of which represent "forbidden" evolutionary trajectories (Merilä and Björklund 2004; see also Schluter 1996). Hence, depending on the direction of deviation of the population mean with respect to the selective optimum, it is possible that this optimum cannot be regained (Merilä and Björklund 2004) due to the constraints imposed by the covariance structure. Our results can be consistent with this being the case in third molars.

INTEGRATION AMONG TEETH OF THE SAME ARCADE

Our analysis of integration among different teeth reveals a general and strong evolutionary integration in the complete postcanine dentition. In both arcades, integration is stronger between molars than between premolars, but significant covariation between premolars and molars reveals no modularization of a premolar and a molar field. These results are not in complete agreement with previous studies of buccolingual and mesiodistal dental dimensions in modern human populations that have revealed that tooth type accounts for the majority of size variance among different teeth (Harris 2003). Similarly, quantitative genetic studies have demonstrated the presence of at least three different modules: an incisor module genetically independent of a postcanine module, and a premolar module that has incomplete pleiotropy with the molar module (Hlusko and Mahaney 2009; Hlusko et al. 2011). Unfortunately, the present analysis of hominin dentition has not included incisors and canines, thus making it difficult to compare these results with other studies of modularity. Nevertheless, our results can be compared with the specific evaluation of postcanine teeth in the aforementioned studies. Even so, it is important to note that the majority of empirical studies on dental integration are based on size variables, whereas the present analysis has evaluated shape characters. This is relevant to this discussion because it has been suggested that size can be more evolutionary labile than shape, the evolution of which is more constrained (Hunt 2007b).

All the teeth coevolving together show a coordinated reduction of their distal areas (see Figs. S11–S21) that can be supporting a new level of modularity proposed using quantitative genetics in the baboon mandibular dentition (Hlusko et al. 2004). This level of modularity would integrate features from different teeth independently of other parts of the crown, thus giving rise to a mesial module and a distal module. This different organization of modularity would be also supported by the relative constancy of mesial cusps size and the high variability of distal cusps size observed in hominoids (Corruccini 1979) and humans (Kondo and Yamada 2003; Takahashi et al. 2007). Jernvall (2000) explained the high frequency of cusp loss in the most distal molars as the consequence of the early termination of cusp morphogenesis, which would give rise to a paedomorphic tooth that does not completely realize its potential cusp pattern. Both time and space restrictions can be involved in this incomplete pattern realization, and they both point to a substantial epigenetic influence on dental complexity and morphology (Townsend et al. 2003). The early formation and calcification of the protocone and, especially, of the paracone would leave little space for distal cusps to develop in shortened modern human jaws (Trinkaus 2003). As for time restrictions, a heterocronic phenomenon of postdisplacement would delay the onset of dental formation in late hominin species, whereas the rate of development and the offset signal would remain the same as in ancestral species (Alberch et al. 1979). This model can explain the structural reduction of the last developing teeth and, in the most extreme cases, their agenesis, due to the late initiation and relative early termination of dental development. Under this model, a higher integration is expected between premolars and distal molars than between first, second, and third molars. The relative stasis of both first molars versus the more random (lower molars) and directional (upper molars) patterns of evolution of distal molars can be related with the shift on molar proportions observed from earlier ($M1 < M2 < M3$) to later hominins ($M1 > M2 > M3$). This alteration on size relationships among molars is explained by Kavanagh et al. (2007) by means of a simple developmental cascade based on the balance between activator and inhibitor signaling molecules (see also Polly 2007). The same model can be used to explain the morphological reduction of second and third molars in late hominin species as a consequence of an increase in inhibition versus activation that is commonly observed in animal-eating species.

Another important result of the present work is the observed stronger integration among mandibular postcanine teeth than among maxillary teeth. Again, this observation is fully compatible with quantitative genetic studies that have reported complete pleiotropy in the development of baboon mandibular series versus incomplete pleiotropic effects in the maxillary

dentition (Hlusko et al. 2004). Similarly, studies analyzing craniofacial and mandibular integration have revealed the existence of different patterns of craniofacial integration in modern humans and African apes (e.g., Ackermann 2002; Polanski and Franciscus 2006), whereas general similarities exist in mandibular integration (see Polanski 2011). The similarities in mandibular integration would be the result of the passive role played by the mandible during hominoid and hominin evolution (Polanski 2011), because mandibular changes would be secondary consequences of the primary changes undergone by the cranium and face (Lieberman et al. 2004). If certain degree of coevolution can be expected between teeth and jaws (see Plavcan and Daegling 2006; Boughner and Hallgrímsson 2008; Cobb and Panagiotopoulou 2011), differences in the strength of integration between the maxillary and mandibular dentitions are not surprising. The lower dentition would remain highly integrated within the more stable environment of the mandible, whereas the upper dentition would respond to strong changes in the cranium and in the face by means of a weaker integration.

These different degrees of integration can be also related to the observed predominant evolutionary modes in the upper and lower dentitions. Mandibular teeth would evince a stronger stasis associated to the relatively stable mandibular morphology and integration (Polanski 2011; see also Bastir et al. 2005). On the contrary, maxillary teeth would show a more neutral pattern of evolution that is clearly consistent with the aforementioned studies demonstrating random factors in the morphological evolution of cranial morphology (Ackermann and Cheverud 2004; Weaver et al. 2007; von Cramon-Taubadel 2009; Betti et al. 2010).

INTEGRATION AMONG ANTAGONIST TEETH

The patterns of integration observed among antagonist teeth consist on strong covariation between first molars (observed both inter- and intraspecifically) that decreases gradually toward first premolars and third molars, with intermediate values corresponding to second premolars and second molars. Functional constraints in the evolution of the dentition prevent teeth from undergoing directional or even random changes (Polly et al. 2005) because cusps of occluding teeth must fit perfectly to maintain a correct occlusion (Evans and Sanson 2003). Hence, a change in a given tooth will need a correlated change in the antagonist to keep their functionality (Polly 2004). This functional integration has a strong influence on dental evolution in spite of the reported partial genetic independency in the development of the upper and lower dentitions (e.g., McCollum and Sharpe 2001; Shimizu et al. 2004; Charles et al. 2009). Several studies analyzing correlations between antagonist teeth have found a high phenotypic integration between antagonists, specially between first molars (Gingerich and Winkler 1979; Szuma 2000; Prevosti and Lamas 2006; Renaud et al. 2009). The present study also reveals the strongest integration, as well as the strongest stasis, in both first molars (Clyde and Gingerich 1994; Wood et al. 2007; Piras et al. 2009). The described pattern of integration matches reasonably well the observed patterns of evolutionary changes in the different dental classes: predominant stasis in both first molars that changes gradually toward certain degree of directional change in lower first premolars and upper third molars, and predominant random change in the other teeth.

The key role of first molars for correct occlusion has been recognized since the beginning of the 20th century, in Angle's classic work (Angle 1899). First molars have a fundamental developmental role because they are the first permanent teeth to erupt. As such, first molars have a significant influence on the position of later erupting teeth and on the vertical distance between the upper and lower jaw. Besides, in the most recent hominin species, first molars are the largest teeth with the strongest anchorage to both jaws, and their position in both arcades makes them suffer the main load during mastication. The central developmental and functional role of first molars is likely to cause a strong integration and a general stasis through the course of hominin evolution. On the contrary, later erupting teeth, especially third molars, have less important roles in occlusion that allow them to evolve in a more independent way from their antagonists. These teeth can then undergo certain degree of directional selection—previously observed in *Australopithecus* evolution (Lockwood et al. 2000)—without compromising occlusion.

Possible causes for the directional trends observed in lower first premolars and upper third molars differ. In the case of premolars, a strong reduction during hominin evolution of the canine-P_3 honing complex—which was likely present in the chimpanzee-hominin last common ancestor (Cobb 2008)—has been documented. Although clear canine-P_3 honing complexes are not present in any of the species included in this study, australopith species have strongly asymmetric premolars (Leonard and Hegmon 1987; Asfaw et al. 1999; Lockwood et al. 2000; White et al. 2006; Gómez-Robles et al. 2008) that can be considered as a remnant of their ancestral sectorial condition. As for third molars, their strong reduction in size (Mizoguchi 1983; Macho and Moggi-Cecchi 1992) and complexity (Williams and Corruccini 2007) in latest *Homo* species has been classically related with facial and mandibular reductions associated with dietary changes (Calcagno and Gibson 1988; Armelagos et al. 1989; Calcagno and Gibson 1991). The directional evolutionary trend observed in third molars can be also the indirect result of the evolutionary pressure delaying the onset of dental formation as a consequence of the extended childhood observed in late hominin species (see Bermúdez de Castro 1989). The lack of a directional trend in the reduction of lower third molars can be tentatively explained by the weaker integration and the more random pattern of reduction of these molars, as explained above.

EVOLUTION, DEVELOPMENT, AND FUNCTION

In spite of the different evolutionary dynamics observed in the different dental classes, the whole postcanine dentition is strongly integrated. This initially counterintuitive result is caused by the maintenance of a significant phylogenetic signal (defined as the degree to which phylogenetic relatedness among taxa is associated with their phenotypic similarity, Klingenberg and Gidaszewski 2010) despite the functional and developmental constraints that have an influence on dental evolution. As serially homologous structures, teeth share large proportions of their genetic and developmental architecture (see Young and Hallgrímsson 2005), so high developmental integration can be assumed by default between different teeth. However, the aforementioned partial genetic independence of the upper and lower dentition points to a parcellation of the whole dentition in a maxillary and mandibular module, probably achieved through changes in the extent of pleiotropic effects (e.g., Cheverud 1996; Wagner 1996; Mitteroecker and Bookstein 2008; Rolian 2009) by means of a differential regulation of common developmental pathways (e.g., Stock 2001; Tucker and Sharpe 2004; Plikus et al. 2005). Genetic integration would be then an initial property of serially homologous systems, not a consequence of functional and developmental integration (Cheverud 1996), and parcellation would evolve in response to functional demands (Young and Hallgrímsson 2005). At the same time, functional constraints would prevent teeth to evolve independently from their antagonists, especially in those positions that are responsible to a stronger degree for a correct occlusion.

Some previous research has revealed selection operating on specific dental features, namely on upper first molar size (DeGusta et al. 2003). Other studies have evaluated the relationship between dental wear and fitness, concluding that "loss of dental capacity appears ultimately to limit fitness" (King et al. 2005). This statement implies that, if the loss of dental functionality is caused by changes in dental morphology, dysfunctional tooth shapes can actually decrease fitness and can be negatively selected. The fundamental role of first molars in keeping a correct occlusion and their indirect influence on facial development (see Ackermann and Krovitz 2002) highlight the importance of a general stability of first molars far beyond their masticatory function. Linkages between dental, facial, and mandibular features can then impact the understanding of selection because stasis can be an indirect result of genetic and/or functional links with other traits under selection (Lande and Arnold 1983). The focus of the present study on dental evolution does not allow for an evaluation of these associations, but it reveals that stabilizing selection can be operating over first molar shape, be it direct or indirect.

Stabilizing selection has been often regarded as a possible cause for the stasis observed in dental morphology due to strong functional constraints (e.g., Clyde and Gingerich 1994; Polly 2004; Wood et al. 2007). It has also been suggested that differences in cusp configuration (specifically, the gain of determine cusps) could be directly involved in the invasion of new adaptive zones (Hunter and Jernvall 1995). Nonetheless, the link of slight mismatchings between occluding teeth and a decreased fitness is not completely clear, especially in increasingly technological hominin populations. An alternative role of stabilizing selection on the morphological stasis of first molars can be involved through its effect on developmental pathways by removing the variants that would give rise to outliers (Polly 1998). This mechanism, frequently invoked to explain reduced morphological variability, is called canalization (Waddington 1942), and it refers to the property of organisms to buffer development against environmental and genetic perturbations (see Hallgrímsson et al. 2002; Willmore et al. 2007). The resultant consistency of phenotypic expression under a variety of conditions that fall under a particular threshold (Willmore et al. 2007) might give rise to the observed morphological stability of first molars without the direct effect of stabilizing selection on dental morphology.

It is generally assumed that, if patterns of covariation differ across several hierarchical levels (among species vs. within species), this will be an evidence of selection overriding constraints (Merilä and Björklund 2004). However, the opposite case where patterns of integration are coincident among and within species (as, in general terms, is observed in this study, see Table 5) is not necessarily the result of evolution being driven by neutral processes and developmental constraints, but selection can be still involved (Merilä and Björklund 2004). Moreover, integration can act as a constraint and as an adaptive feature at the same time in certain cases because significant degrees of covariation among the elements of a homologous series can constrain their evolution, but also facilitate the emergence of new functionalities (see Rolian et al. 2010). Our identification of heterogeneous patterns of integration and evolutionary dynamics in different areas of the dentition points to the influence on dental evolution of both developmental constraints and selection related to functional factors.

Conclusions

The results of this study suggest a scenario for the evolution of morphological integration and modularity in the mammalian dentition, based on the specific example of hominin dentition. As a note of caution to this generalization, it should be recognized that the morphological variation and temporal span represented by hominins is small compared with mammal variation. Nonetheless, some of the patterns we recognize may be pervasive across heterodont organisms (Hlusko et al. 2011), as observed in the general agreement of the hominin and modern human patterns of covariation, and also in the consistency of this study and evaluations of integration and evolutionary modes in other primates, and some

rodents and carnivores. Our model extends the four-level hierarchical organization of morphological integration suggested by Cheverud (1996) to accommodate the serially homologous nature of the dentition. The elements of the dentition (serial homologues) would be genetically and developmentally integrated in the ancestral condition. Different evolutionary and developmental constraints in the maxilla and the mandible would have given rise to an upper and a lower module through the up- or down-regulation of their common developmental pathways. This modularization is empirically demonstrated by the relative independence in the development of the maxillary and mandibular dentitions. This process would favor integration within each jaw as a whole, with some independence between them. Continued evolutionary integration between the jaws in response to functional demands of occlusion could be achieved through at least two different processes. First, functional constraints would expose those teeth with the most important role in occlusion to strong stabilizing selection, whereas teeth with minor roles would be able to undergo slight directional trends in response to different selective pressures without compromising occlusion and related processes. This would result in submodules within the dentition that cross the mandibular and maxillary modules. Second, phenotypic variability in the functionally most relevant elements of the dentition might be reduced through canalization. The inherent developmental origin of canalizing mechanisms would lead to an evolutionary integration via a genetic and/or developmental integration that would be a primary property of serially homologous systems.

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Supporting Information

The following supporting information is available for this article:

Figure S1. Conformation of landmarks and semilandmarks used in the morphological analysis of upper first premolars.
Figure S2. Conformation of landmarks and semilandmarks used in the morphological analysis of upper second premolars.
Figure S3. Conformation of landmarks and semilandmarks used in the morphological analysis of upper first molars.
Figure S4. Conformation of landmarks and semilandmarks used in the morphological analysis of upper second molars.
Figure S5. Conformation of landmarks and semilandmarks used in the morphological analysis of upper third molars.
Figure S6. Conformation of landmarks and semilandmarks used in the morphological analysis of lower first premolars.
Figure S7. Conformation of landmarks and semilandmarks used in the morphological analysis of lower second premolars.
Figure S8. Conformation of landmarks and semilandmarks used in the morphological analysis of lower first molars.
Figure S9. Conformation of landmarks and semilandmarks used in the morphological analysis of lower second molars.
Figure S10. Conformation of landmarks and semilandmarks used in the morphological analysis of lower third molars.
Figure S11. Morphological covariation between upper premolars and molars as corresponding to the first singular axes in the interspecific comparison.
Figure S12. Morphological covariation between upper molars as corresponding to the first singular axes in the interspecific comparison.
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Figure S17. Morphological covariation between upper premolars and molars as corresponding to the first singular axes in the intraspecific comparison.
Figure S18. Morphological covariation between upper molars as corresponding to the first singular axes in the intraspecific comparison.
Figure S19. Morphological covariation between lower premolars as corresponding to the first singular axes in the intraspecific comparison.
Figure S20. Morphological covariation between lower premolars and molars as corresponding to the first singular axes in the intraspecific comparison.
Figure S21. Morphological covariation between antagonist teeth as corresponding to the first singular axes in the intraspecific comparison.

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